

Challenge 2: Estimating State of Health under Real-World Operation

1. The Task

Grid-forming battery storage is dispatched, warranted and insured on the basis of a quantity nobody in the field can measure directly: the state of health of its packs. Capacity tests are a laboratory luxury; in operation, the battery management system sees only current, cell voltages and temperatures under a load that changes every minute – and yet it has to know how much capacity is left, to protect the asset, to plan replacements and to keep the guarantee honest. Whether a model can track the health of a real pack from what a BMS sees, at the accuracy a product decision requires, is one of the most practical open questions in battery analytics – and the question this challenge puts to you.

Your task: take the operating data of a real 4-cell pack built from our prismatic 102 Ah LFP cells – current, cell voltages, temperatures and BMS-style Ah counters under a simulated storage-application profile at 25 and 45 °C – and estimate the pack's SOH at every monthly checkup date, using only the data recorded before that date. You receive a single capacity test at beginning of life; all later checkups are held out and used for scoring. As in Challenge 1, the SOH throughout the challenge is the full available (C/20) discharge capacity divided by the nominal capacity – 102 Ah for this pack of four 102 Ah cells in series. The beginning-of-life checkup therefore reads 100.41 Ah = 98.44 % SOH, not 100 %.

1. **Access and process.** You receive the operating data of the pack (13 segments, January to September 2025, 10-s sampling) and the beginning-of-life checkup CK0 with its complete C/20 discharge curve. Get familiar with the dataset. Real lab data means real imperfections: one operating segment (19 February to 11 March) is not provided, there are recording gaps inside segments (the Ah counters stay correct across both), duplicate timestamps mark step transitions, and the load is an application profile, not a cycling protocol.
2. **Use open data where it helps (optional).** Open-source battery datasets may support your approach – to pre-train a model, to calibrate a physics-based one, or to test its transferability. This is an option, not a requirement; a strong estimator can be built from the released data alone. Any external dataset must satisfy the rules in section 2.1.
3. **Build the estimator on the target pack.** Adapt your approach to the pack using only the provided data: the operating signals and CK0. Physics-based, data-driven or hybrid – your choice.
4. **Estimate causally.** For every checkup date the framework hands your model only the data recorded before it. Return the pack SOH in percent of the nominal 102 Ah. A reference baseline (partial-charge coulomb counting anchored at CK0) is provided and defines the score of 1.0.
5. **Do more with less (bonus).** Every submission is additionally evaluated in a reduced-budget run with only part of the operating data (exact budget in the Submission & Validation Guide). Models that keep their accuracy with less data earn a data-efficiency bonus on top of their score.
6. **Package and submit as often as you like.** Standardized command-line interface and output CSV (identical to Challenge 1: `run_model.py --model train / test`); a public leaderboard is updated weekly; only submissions that run successfully through the automated pipeline are scored.

Given: (a) the operating data of the 4-cell 102 Ah LFP pack over eight months, (b) the beginning-of-life checkup CK0 (capacity and discharge curve), (c) the dates of all checkups CK0–CK7.

Wanted: one SOH estimate (percent of the nominal 102 Ah) for every checkup date CK1–CK7 listed in section 2.3, each computed from the data recorded before that date. Estimates are evaluated against the hidden C/20 capacities measured at those checkups.

Relevance of the problem: the flat voltage plateau of LFP makes state estimation from operating data genuinely hard – a voltage threshold is an unreliable SOC anchor, imbalance between the cells grows over life, and the temperature changes the response. A naive estimator sees only part of the fade; a good one turns the response to a known, repeating load into a measurement of health. The pack studied here is the building block of a real product, and the checkups it was scored against are exactly the measurements a BMS never gets.

2. Data

2.1 Open-source pre-training data

All datasets used or provided in Challenge 1 may be used again, and further open-source datasets are welcome. Particularly useful are pack- and system-level datasets of large LFP cells under real or realistic load, for example the field dataset

published by TU Darmstadt [1]: 28 LFP battery systems with eight prismatic cells in series (about 160 Ah), cell voltages, pack current and temperatures over up to five years of operation. Any dataset must satisfy the following two rules: (1) documented with citation and license in the final report; (2) reproducible by the pipeline (download scripts or declared cached copies). Private or proprietary external data is not allowed. Performance on open data is not scored.

[1] – Schaeffer, Joachim, et al. "Gaussian process-based online health monitoring and fault analysis of lithium-ion battery systems from field data." Cell Reports Physical Science (2024): 102258. Data: <https://doi.org/10.5281/zenodo.13715694> (CC BY-NC 4.0).

2.2 The pack operating dataset (provided as download)

A 4-cell series pack (4 × 102 Ah LFP, 12.8 V nominal) was operated for eight months with a simulated storage-application profile: a daily charge/discharge pattern with realistic variability and, roughly every five days, a full CCCV charge to 14.0 V followed by a 30-min 20 A discharge pulse. Operation was recorded in segments alternating between 25 °C (short) and 45 °C (long), about 156 equivalent full cycles in total (13 of the 14 segments are provided); every month the pack was taken out for a reference performance test with a C/20 capacity measurement (CK0–CK7).

File	Content (one row per ...)
operation/segment_NN_TTdegC.csv.gz	10 s record: timestamp, elapsed_s, segment, chamber_temperature_C, current_A (+ charge / - discharge), charge_Ah_cum and discharge_Ah_cum (BMS-style counters, cyclers accounting, never reset), cell1_V..cell4_V, pack_voltage_V, temp_mean/min/max_C. One file per segment; segment 3 (19 Feb – 11 Mar) is not provided.
checkups/checkup_capacities_released.csv	CK0: date, C/20 capacity (Ah), SOH (% of 102 Ah) - the only capacity value released.
checkups/evaluation_points.csv	all checkup dates CK0..CK7 with a released flag; one SOH estimate is required per date.
checkups/CK0_reference_discharge.csv.gz	the beginning-of-life C/20 discharge curve (5.1 A to 11.2 V): pack and cell voltages vs discharged Ah.

Protocol: simulated application profile (about 0.8 equivalent full cycles per day, currents within ±20 A) at fixed chamber temperature; 10 s sampling; pack voltage window 11.1–14.1 V. Checkups: C/20 discharge at 5.1 A to 11.2 V after a full charge; their throughput is not part of the operating data or the Ah counters.

2.3 Evaluation dates – the SOH estimates you must deliver

The goal of the challenge is to estimate the pack SOH at each of the following dates. Your model returns one SOH value per date; the dates are known to you, the capacities measured there are not. Please note you only received the beginning-of-life checkup CK0 – the capacities of the seven later checkups are withheld and used only for evaluating your model.

Checkup	Date	Your task at this date
CK0	2025-01-14	anchor – released: 100.41 Ah = 98.44 % SOH (returned, not scored)
CK1	2025-02-08	estimate the pack SOH – scored against the hidden capacity
CK2	2025-03-11	estimate the pack SOH – scored against the hidden capacity
CK3	2025-04-14	estimate the pack SOH – scored against the hidden capacity
CK4	2025-05-16	estimate the pack SOH – scored against the hidden capacity
CK5	2025-06-16	estimate the pack SOH – scored against the hidden capacity
CK6	2025-07-25	estimate the pack SOH – scored against the hidden capacity
CK7	2025-08-28	estimate the pack SOH – scored against the hidden capacity

Timeline, submission platform and deadline: see the Submission & Validation Guide shared with this document.